The Architecture of Insight: A Reduced-Model Hypothesis for Representational Flexibility and Insight Capacity

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Abstract

Insight—the sudden restructuring of a problem representation that yields a novel solution—has been associated with state-dependent factors including affect and attentional conditions. We introduce the **Representational Flexibility Theory of Insight Capacity** (RFTIC), a hypothesis that separates such state variables from a putative structural property of semantic connectivity. Within a reduced continuous-Hopfield model (Hopfield, 1982, 1984), orthonormal domain patterns and hypothesis (A1) imply the exact raw-eigenvalue shift $\lambda_W(\varepsilon) = \lambda_{W0} - \varepsilon$. With $\mu(\varepsilon) = \lambda_W(\varepsilon) - 1 = \mu_0 - \varepsilon$, $\mu_0 = \lambda_{W0} - 1 > 0$, and the additional transverse-gap hypothesis (A2), partial minimization of the Hopfield energy gives an asymptotic normal form near $\mu \to 0^+$. Its quartic truncation has coefficient $g = \sum_i (v_i^*)^4$, minima $x_\pm = \pm\sqrt{3\mu/g}$, barrier $\Delta E(\varepsilon) = 3(\mu_0 - \varepsilon)^2/(4g)$, and critical value $\varepsilon_{\text{crit}} = \mu_0 = \lambda_{W0} - 1$. The order parameter vanishes with mean-field exponent $\beta = 1/2$ as the two wells merge; above the critical point the reduced potential is single-welled, so a basin-to-basin escape rate is no longer the relevant observable. We validate the internal numerical implementation of the canonical normal-form dynamics using 16 values of $r = \Delta E/T$ and 6,000 trajectories per value. At moderate-to-high barriers, the stabilized conditional hazard agrees closely with the inverse one-dimensional MFPT. At lower barriers, the two estimands differ. Kramers' formula progressively approaches the inverse MFPT at high barriers. These results establish consistency of the reduced model only; they do not validate its proposed cognitive, clinical, or artificial-intelligence interpretations.

Keywords: insight; representational flexibility; Kramers escape rate; Hopfield network; first-passage time; reduced stochastic model

1. Introduction

1.1. State-Dependent Findings and an Explanatory Gap

Insight performance has been associated with several task- and state-dependent factors, including positive affect and patterns of attentional or executive engagement (Subramaniam et al., 2009; Kounios and Beeman, 2014). The literature does not establish a single universal state of reduced analytical control, and effects must be interpreted in relation to the specific task and manipulation. In verbal insight tasks, Jung-Beeman et al. (2004) observed greater right anterior superior temporal activity for insight than noninsight solutions and a right-temporal gamma-band increase beginning approximately 300 milliseconds before the button-press response. These findings do not by themselves demonstrate global prefrontal suppression, Default Mode Network activation, or that the gamma increase precedes the first conscious awareness of the solution.

Performance and reported Aha experiences vary across tasks and individuals (Danek et al., 2014; Webb et al., 2018). Established predictors are also important: Chuderski and Jastrzębski (2018) found that working-memory capacity and reasoning were strongly related to insight performance. RFTIC therefore proposes an additional candidate construct rather than assuming an already isolated residual factor. Its incremental predictive validity must be tested against working memory, reasoning, prior knowledge, task characteristics, and momentary state.

We term this hypothesized individual-level construct *representational flexibility* (FR): the putative capacity to encode the same problem under multiple representations and to transition between them. The central hypothesis is that FR may contribute to *insight capacity*—as distinct from insight probability in a given moment. Whether FR is stable, trainable, measurable independently of behavioral

outcomes, or separable from established cognitive predictors remains an empirical question.

1.2. The Representational Change Theory and Its Limits

The theoretical framework most directly relevant to our proposal is Ohlsson's Representational Change Theory (RCT; Ohlsson, 1992, 2011). RCT proposes that insight occurs through two mechanisms: *constraint relaxation* (the softening of self-imposed problem framings) and *chunk decomposition* (the disaggregation of perceptual or conceptual units that block alternative encodings). These mechanisms were tested in the original matchstick-arithmetic experiments of Knoblich et al. (1999), with related processes subsequently examined in other insight tasks (Knoblich et al., 2001; Öllinger et al., 2008, 2014; Danek et al., 2020). RCT is thus a prominent process-level account of insight rather than an exclusive or complete account of every insight phenomenon.

RCT describes representational operations associated with the transition from impasse to solution. RFTIC introduces a complementary level of analysis concerned with how a model's transition propensity might vary across systems or individuals. The distinction between a theory of an event and a theory of a proposed capacity is our conceptual framing, not a claim that RCT explicitly excludes individual differences.

This is not a criticism of the theory's internal logic but a recognition of a different level of analysis. The relationship is analogous to that between the kinematics of chemical reactions—which describes bond-breaking and bond-forming sequences—and transition-state theory, which predicts *rates* as a function of energy barriers and fluctuations. RCT motivates the representational operations; RFTIC proposes a statistical-mechanical layer in which a specified perturbation changes a reduced transition barrier. Whether this formalism explains stable human differences in insight is an open empirical question.

1.3. State-Based Accounts and the Friston Constraint

A parallel literature develops the Free Energy Principle and active-inference accounts of perception, action, and, in specific formal models, curiosity and insight (Friston, 2010; Friston et al., 2017). RFTIC uses a distinct stochastic model: active-inference precision and the RFTIC noise parameter T are not established as mathematically or empirically identical, and no cited evidence shows that dopamine controls RFTIC-like exploration.

RFTIC asks a complementary structural question: whether prior connection weights retain training-induced changes that affect later representational transitions. In the reduced stochastic model, T controls noise amplitude while ε changes the potential. Treating these mathematical parameters as neural precision and learned semantic connectivity is an interpretive hypothesis, and their empirical independence has not been established.

1.4. The Trainability Gap

A third line of evidence concerns procedures that can facilitate performance on insight-related tasks. In controlled studies, explicit representational-change procedures improved performance on previously untrained verbal insight problems (Patrick and Ahmed, 2014; Patrick et al., 2015); Patrick et al. (2015) also observed an advantage in an independent trained group tested after 48 hours. This is within-class procedural generalization, not a longitudinal retention curve, far transfer, stable trait change, or evidence of altered neural connectivity. Analogical comparison can facilitate access to structurally related solutions (Gick and Holyoak, 1983; Kurtz and Loewenstein, 2007), and brief mindfulness manipulations have also been studied with insight tasks (Ostafin and Kassman, 2012). These findings motivate candidate interventions but do not establish a single latent FR variable or a common biological mechanism.

RFTIC proposes an organizing framework. The six components in Section 2 organize candidate interventions by their hypothesized effects on W, trajectories, or T. Existing training studies motivate these candidates but do not establish a single latent variable or the specific network changes assumed by the model.

1.5. The Present Paper

We proceed as follows. Section 2 defines FR as a hypothesis and organizes six candidate components with proposed model-level mappings. Section 3 states the assumptions and derives the chain $W(\varepsilon) \to \lambda_W(\varepsilon) \to \mu(\varepsilon) \to U(x;\varepsilon) \to \Delta E(\varepsilon) \to k \to P_\tau$. Section 4 compares Euler–Maruyama simulation with an exact one-dimensional first-passage calculation and the Kramers asymptotic limit. Section 5 separates predictions of this reduced model from speculative cognitive, clinical, and artificial-intelligence implications.

The argument of this paper rests on a single conceptual inversion: the literature has asked, *what states facilitate insight?* We ask, *what structural properties might contribute to representational transition capacity beyond transient state-dependent variation?* The second motivates a complementary level of analysis focused on capacity rather than a single insight event.

2. The RFTIC Framework

2.1. Definition and Scope

Representational flexibility (FR) is defined within RFTIC as a hypothesized individual-level construct concerning connectivity across representational domains. Participant-specific semantic association networks can in principle be estimated from suitable behavioral data, and group-level topological differences have been studied in related creativity research (Kenett et al., 2014; Zemla and Austerweil, 2018). We therefore propose that within- and across-domain network structure may provide candidate operationalizations of FR. Corpus-trained Word2Vec, GloVe, and BERT representations provide exploratory geometric spaces (Mikolov et al., 2013; Pennington et al., 2014; Devlin et al., 2019), but they do not measure a person's mind. Cosine distance is not a validated FR measure, and semantic embedding geometry is not psychological FR by definition.

In the reduced model, the hypothesized structural contribution is parameterized by a nonnegative scalar ε, designated the **Topological Flexibility Index (TFI)**. TFI is a proposed model parameter rather than a validated psychometric variable, and ε has not been empirically calibrated from any network or embedding metric. The double-well regime exists only for $0 \leq \varepsilon < \varepsilon_{\text{crit}}$, where $\varepsilon_{\text{crit}} = \mu_0 = \lambda_{W0} - 1$ is derived in Section 3; this mathematical interval is not an empirically calibrated cognitive scale.

This proposed measurement strategy is intended to avoid the central tautology that threatens any capacity-based theory of insight: if FR were defined as "the tendency to change representations successfully," and insight as "successful representational change," the argument would be circular. By grounding FR in participant-level network descriptors estimated independently of the target problem-solving episode, future studies could test reliability, construct validity, and incremental prediction rather than defining the construct by the outcome it is intended to explain.

2.2. Six Components of Representational Flexibility

We propose six candidate components and intervention targets that may influence representational flexibility through distinct psychological processes. Their assignments to W, search trajectories, and T are explanatory modeling choices, not findings of the cited experiments.

Component 1 — Inconsistency Detection. Inconsistency-evaluation procedures have been tested as ways to cue re-encoding during verbal insight problem solving (Patrick and Ahmed, 2014; Patrick et al., 2015). We hypothesize, without direct neural evidence, that repeated use of such a procedure could be represented by a selected connectivity update in W_{AB}, by analogy with Hebbian learning (Hebb, 1949).

Component 2 — Cross-Domain Analogical Transfer. Research on analogical comparison and transfer motivates the hypothesis that practice in cross-domain mappings could change effective associative structure (Gick and Holyoak, 1983; Kurtz and Loewenstein, 2007). The modeled update $\tilde{W}_{AB} = \varepsilon(\psi_A\psi_B^T + \psi_B\psi_A^T)$ is the perturbation analyzed in Theorem 1; it is an RFTIC modeling assumption, not a connectivity equation inferred from those experiments.

Component 3 — Tolerance of Impasse. A candidate capacity to sustain representational ambiguity without premature closure. Direct intervention evidence is not established here. In the model analogy, Component 3 changes the available observation or search time rather than W itself.

Component 4 — Metacognition of Impasse. A proposed capacity to distinguish representational impasse from analytic difficulty. Foundational metacognitive accounts motivate monitoring and control concepts (Flavell, 1979; Nelson and Narens, 1990), but they do not establish an RFTIC-specific impasse classifier. Such classification is hypothesized to direct exploratory effort toward representational change when impasse is genuine.

Component 5 — Type 1 Disinhibition. Transient relaxation of habitual analytic monitoring is considered as a candidate intervention target. The Type 1 terminology is descriptive (Kahneman, 2011), and mindfulness research provides a task-level precedent rather than evidence of voluntary prefrontal suppression (Ostafin and Kassman, 2012). Within RFTIC, Component 5 is represented as a transient change in T rather than in W; this theoretical mapping does not identify T with neural precision.

Component 6 — Chunk Decomposition. The trained capacity to disaggregate perceptual or conceptual units fused through expertise or habit (Knoblich et al., 1999). The model maps chunk decomposition to changes in the intra-domain blocks W_{AA} and W_{BB} and hence to the raw baseline eigenvalue λ_{W0} and net coefficient $\mu_0 = \lambda_{W0} - 1$. Within the truncated normal form, changing μ_0 changes both $\Delta E = 3(\mu_0 - \varepsilon)^2/(4g)$ and the critical point; the proposed neural mapping has not been demonstrated, and no cited study shows a stable training-induced modification of neural W.

2.3. Summary: Coordinated Structural Modification

Table 1 summarizes the proposed components, candidate intervention targets, and hypothesized model-level mappings.

In the proposed mapping, Components 1, 2, and 6 act on structural parameters; Components 3 and 4 act on search trajectories; and Component 5 acts on T. These categories are modeling hypotheses rather than established biological mechanisms.

3. Mathematical Formulation

3.1. Network, transition mode, and training perturbation

Let $S \in \mathbb{R}^N$ be a continuous activation vector and let the symmetric matrix $W \in \mathbb{R}^{N\times N}$ encode the semantic connectivity. The bilinear Hopfield energy is

$$E(S) = -\tfrac{1}{2} S^T W S, \tag{1}$$

with the domain partition

$$W = \begin{pmatrix} W_{AA} & W_{AB} \\ W_{BA} & W_{BB} \end{pmatrix}, \qquad W_{AB} = W_{BA}^T. \tag{2}$$

RFTIC denotes the proposed psychological capacity by representational flexibility (FR). Its model parameter, the Topological Flexibility Index (TFI), is $\varepsilon \geq 0$; FR and ε are therefore not interchangeable quantities. Let the two domain patterns satisfy

$$\|\psi_A\| = \|\psi_B\| = 1, \qquad \psi_A^T \psi_B = 0, \tag{3}$$

and define the normalized contrastive direction, its symmetric companion, the transverse space, and the reduced coordinate by

$$\begin{aligned} v^* &= \frac{\psi_A - \psi_B}{\sqrt{2}}, & u &= \frac{\psi_A + \psi_B}{\sqrt{2}}, \\ V_\perp &= \{w : w^T v^* = 0\}, & x &= v^{*T} S. \end{aligned} \tag{4}$$

ψ_A and ψ_B define the contrastive direction used to construct the transition mode, whereas the minima of the reduced normal form represent effective states along that coordinate rather than the original pattern vectors themselves. The cross-domain Hebbian perturbation is

$$\begin{aligned} \tilde{W}_{AB}(\varepsilon) &= \varepsilon(\psi_A \psi_B^T + \psi_B \psi_A^T), \\ W(\varepsilon) &= W_0 + \tilde{W}_{AB}(\varepsilon). \end{aligned} \tag{5}$$

Lemma 1 (Action of the training term)**.** *Under* (3)*,* $\tilde{W}_{AB}(\varepsilon) v^* = -\varepsilon v^*$.

Proof. Writing $a = \psi_A$ and $b = \psi_B$,

$$\begin{aligned} (ab^T + ba^T)\frac{a-b}{\sqrt{2}} &= \frac{a[b^T(a-b)] + b[a^T(a-b)]}{\sqrt{2}} \\ &= \frac{-a+b}{\sqrt{2}} = -v^*. \end{aligned}$$

Multiplication by ε proves the statement. □

The reduction requires the following explicit baseline hypothesis:

$$\begin{aligned} &W_0 v^* = \lambda_{W0} v^*, \qquad \lambda_{W0} = v^{*T} W_0 v^*, \\ &\quad \mu_0 := \lambda_{W0} - 1 > 0. \end{aligned} \tag{A1}$$

Hypothesis (A1) concerns only v^*; it does not require ψ_A and ψ_B to be individual eigenvectors or W_0 to be isotropic on their two-dimensional span.

Theorem 1 (Exact shift along the transition mode)**.** *Under* (3) *and (A1),* v^* *remains an eigenvector of* $W(\varepsilon)$ *and*

$$\begin{aligned} \boxed{\lambda_W(\varepsilon) = \lambda_{W0} - \varepsilon}, \\ \boxed{\mu(\varepsilon) := \lambda_W(\varepsilon) - 1 = \mu_0 - \varepsilon}. \end{aligned} \tag{6}$$

Proof. By (A1) and Lemma 1, $W(\varepsilon)v^* = W_0 v^* + \tilde{W}_{AB}(\varepsilon)v^* = (\lambda_{W0} - \varepsilon)v^*$. □

Thus the coefficient of ε is fixed at one rather than fitted. Without (A1), $v^{*T} W(\varepsilon) v^*$ is a Rayleigh quotient, not necessarily an eigenvalue, and the neglected coupling to directions orthogonal to v^* becomes an additional approximation. The raw eigenvalue λ_W and the effective quadratic coefficient $\mu = \lambda_W - 1$ are distinct; the unit shift comes from the quadratic saturation term in the Hopfield energy.

The one-dimensional reduction additionally requires a uniform transverse gap on the interval $I \subseteq [0, \varepsilon_{\text{crit}})$ under consideration:

$$\sup_{\varepsilon \in I} \lambda_{\max}\Big(W(\varepsilon)\big|_{V_\perp}\Big) \leq 1 - \gamma, \qquad \gamma > 0. \tag{A2}$$

Hypothesis (A2) is additional to (A1). It supplies transverse stability and convexity sufficient for the reduction, but it has not been established for real semantic networks. A two-dimensional model for cases in which the symmetric mode also becomes critical is outside the present scope.

Lemma 2 (Transverse minimizer)**.** *Under (A1) and (A2), the Hopfield energy is uniformly strongly convex on each transverse slice. Near* $x = 0$ *there is a unique smooth minimizer* $y^*(x) \in V_\perp$*. It satisfies* $y^*(0) = 0$ *and* $y^*(-x) = -y^*(x)$*, with* $y^*(x) = O(x^3)$.

The result follows from $D(S) \succeq I$ in the full Hessian $H(S) = -W + D(S)$, the gap in (A2), and the implicit function theorem. A proof and the explicit slaving coefficient are given in Supplementary Note S1.

3.2. Reduced potential, barrier, and critical point

Hopfield's graded-response model supplies a general inverse-transfer Lyapunov energy (Hopfield,

Table 1: Proposed RFTIC components, candidate intervention targets, and hypothesized model-level mappings.

Component	**Target**	**Candidate mechanism**	**Hypothesized model effect**
1 — Inconsistency Det.	W_{AB}	Sparse Hebbian	$\uparrow \varepsilon$ (targeted)
2 — Analogical Transfer	W_{AB}	Structured Hebbian	$\uparrow \varepsilon$ (aligned)
3 — Impasse Tolerance	Trajectory	Dwell-time extension	Longer search window τ
4 — Metacognition	Trajectory	Routing optimization	Effective search window τ
5 — Type 1 Disinhibition	T	Monitoring shift	$\uparrow T$ (transient)
6 — Chunk Decomp.	W_{AA}, W_{BB}	Intra-domain weakening	$\downarrow \mu_0 \rightarrow \downarrow \Delta E$

1984). For a tanh transfer function with the gain/resistance normalization adopted here, symmetric weights, state/output identification, and zero external input, that general energy reduces to the following normalized special case:

$$E_{\text{Hopfield}}(S) = -\tfrac{1}{2} S^T W S + \sum_i \int_0^{S_i} \tanh^{-1}(s)\, ds. \tag{7}$$

The local expansion

$$\int_0^z \tanh^{-1}(s)\, ds = \frac{z^2}{2} + \frac{z^4}{12} + \frac{z^6}{30} + \mathcal{O}(z^8) \tag{8}$$

gives, on the line $S = xv^*$,

$$\begin{aligned} \mathcal{S}(x) &:= \sum_i \int_0^{x v_i^*} \tanh^{-1}(s)\, ds, \\ \mathcal{S}(x) &= \frac{x^2}{2} + \frac{g}{12} x^4 + \frac{h}{30} x^6 + \mathcal{O}(x^8), \\ g &:= \sum_i (v_i^*)^4, \qquad h := \sum_i (v_i^*)^6. \end{aligned} \tag{9}$$

The $x^2/2$ contribution is the unit-leak shift: the quadratic coefficient is $\mu = \lambda_W - 1$, not the raw eigenvalue λ_W. For normalized v^*, g is its inverse participation ratio and satisfies $1/N \le g \le 1$.

At Level A, define the reduced energy by partial minimization over the stable transverse directions,

$$U(x;\varepsilon) := \min_{y \in V_\perp} E_{\text{Hopfield}}(xv^* + y). \tag{10}$$

Proposition 1 (Asymptotic reduced normal form)**.** *Under (A1)–(A2), for sufficiently small $|x|$,*

$$U(x;\varepsilon) = \frac{g}{12} x^4 - \frac{\mu(\varepsilon)}{2} x^2 + c_6 x^6 + \mathcal{O}(x^8). \tag{11}$$

The transverse minimization first changes the line-restricted energy at order x^6; in particular, it does not alter the quadratic or quartic coefficients.

The proof sketch inserts the slaving relation $y^*(x) = O(x^3)$ from Lemma 2 into the energy expansion and minimizes the resulting transverse quadratic form. The formula for c_6, its bounds, and the full derivation are given in Supplementary Note S2. Thus (11) is an asymptotic reduction near the bifurcation, not an exact finite-μ identity.

At Level B, the paper adopts the quartic truncation

$$U_{\text{nf}}(x;\varepsilon) = \frac{g}{12} x^4 - \frac{\mu(\varepsilon)}{2} x^2 \tag{12}$$

as the normal-form model for the barrier, Kramers rate, first-passage calculation, and simulation. For $\mu > 0$,

$$U'_{\text{nf}}(x) = x\left(\frac{g}{3} x^2 - \mu\right) = 0, \qquad x_\pm = \pm\sqrt{\frac{3\mu}{g}}, \tag{13}$$

where $x_\pm$ are minima and $x = 0$ is the saddle. Direct substitution gives

$$\begin{aligned} U_{\text{nf}}(x_\pm) &= -\frac{3\mu^2}{4g}, \\ \Delta E = U_{\text{nf}}(0) - U_{\text{nf}}(x_\pm) &= \frac{3\mu^2}{4g}. \end{aligned} \tag{14}$$

Combining (6) and (14) gives the Level-B barrier law. The interval displayed below identifies the double-well regime of the quartic model. Its accuracy as an approximation to the Hopfield barrier is asymptotic near $\varepsilon \to \mu_0^-$:

$$\boxed{\Delta E(\varepsilon) = \frac{3(\mu_0 - \varepsilon)^2}{4g}}, \qquad 0 \le \varepsilon < \mu_0, \tag{15}$$

The deterministic critical point is

$$\boxed{\varepsilon_{\text{crit}} = \mu_0 = \lambda_{W0} - 1}. \tag{16}$$

The order parameter is the well separation

$$M(\varepsilon) = x_+ - x_- = 2\sqrt{\frac{3(\mu_0 - \varepsilon)}{g}}. \tag{17}$$

Hence $M \propto (\varepsilon_{\text{crit}} - \varepsilon)^{1/2}$ and the reduced mean-field model has $\beta = 1/2$. This exponent coincides with the mean-field value from Landau theory for a continuous transition in a quartic–quadratic potential (Landau and Lifshitz, 1980). Without a renormalization-group derivation, this coincidence does not establish that RFTIC shares a universality class with any physical system.

As $\varepsilon \to \varepsilon_{\mathrm{crit}}^{-}$, the two minima merge at $x = 0$. At the critical value, $U_{\mathrm{nf}}(x) = gx^4/12$ has a single degenerate quartic minimum. For $\varepsilon > \varepsilon_{\mathrm{crit}}$, the positive quadratic and quartic terms form a single well at the origin, representing the quiescent state along the contrastive coordinate. In this regime the normal form has no two distinct basins along that coordinate, so a basin-to-basin escape rate is undefined rather than zero. The evenness $U(-x) = U(x)$ follows from the global sign symmetry $E_{\mathrm{Hopfield}}(-S) = E_{\mathrm{Hopfield}}(S)$; it is not an independent double-well assumption or an exchange symmetry between ψ_A and ψ_B.

3.3. Kramers rate and finite-window probability

Consider the overdamped Langevin dynamics with friction set to one,

$$dx = -U'(x)\,dt + \sqrt{2T}\,dW_t. \tag{18}$$

Here x is the physical reduced coordinate and the mobility is set to one. For $\mu > 0$, $U''_{\mathrm{nf}}(0) = -\mu$ and $U''_{\mathrm{nf}}(x_\pm) = 2\mu$. Kramers' historical work provides the foundation for activated barrier escape (Kramers, 1940); the expression used here is the corresponding modern overdamped Eyring–Kramers specialization for a committed basin-to-basin transition:

$$\begin{aligned} k(\varepsilon, T) &= \frac{\sqrt{U''(x_-)|U''(0)|}}{2\pi} e^{-\Delta E/T} \\ &= \frac{\mu(\varepsilon)}{\sqrt{2}\,\pi} \exp\left[-\frac{3\mu(\varepsilon)^2}{4gT}\right]. \end{aligned} \tag{19}$$

The prefactor is independent of T and g; those quantities enter the exponential through the barrier ratio. The quantity k has units of inverse time and is not a probability. If the post-transient escape process is approximately Poisson over an observation window τ, the comparable finite-window probability is

$$\boxed{P_\tau(\varepsilon, T) = 1 - \exp[-k(\varepsilon, T)\tau]}. \tag{20}$$

3.4. Dimensionless form and observables

For $\mu > 0$, introduce the canonically rescaled coordinate and time $y = \sqrt{g/\mu}\,x$ and $s = \mu t$. Distinguish the physical noise T from the effective canonical noise

$$\theta := \frac{gT}{\mu^2} = \frac{3}{4r}, \qquad r = \frac{\Delta E}{T} = \frac{3\mu^2}{4gT}. \tag{21}$$

Equation (18) becomes

$$dy = -\left(\frac{y^3}{3} - y\right) ds + \sqrt{\frac{3}{2r}}\,dW_s. \tag{22}$$

Thus the dimensionless dynamics depend only on the invariant barrier ratio r: rescaling the coordinate moves g into the effective noise but does not change $\Delta E/T$. With $\kappa = k/\mu$, the committed Kramers prediction is

$$\boxed{\kappa_K(r) = \frac{1}{\sqrt{2}\,\pi} e^{-r}}. \tag{23}$$

Kramers' formula is asymptotic as $r \to \infty$; $r = 1$ is not a sharp validity boundary.

Two stopping observables must be distinguished. Starting at $x_0 = x_- = -x_+$,

$$\begin{aligned} \tau_{\mathrm{hit}} &= \inf\{t : x(t) \geq 0\}, \\ \tau_{\mathrm{commit}} &= \inf\{t : x(t) \geq x_+\}. \end{aligned} \tag{24}$$

For boundaries $a = -x_+$ and $b = x_+$, the one-dimensional committor is

$$q(x_0) = \frac{\int_a^{x_0} e^{U(y)/T}\,dy}{\int_a^b e^{U(y)/T}\,dy}, \tag{25}$$

so the evenness inherited from $E(-S) = E(S)$ gives $q(0) = 1/2$: reaching the saddle is not yet a committed transition. The half-Gaussian saddle integral implies

$$k_{\mathrm{hit}}/k_{\mathrm{commit}} \longrightarrow 2 \quad \text{as } r \to \infty. \tag{26}$$

The old intermediate threshold measured a different observable. Once the absorbing boundary is the mirror well x_+, no reactive-flux or Grote–Hynes correction is required for this overdamped one-dimensional comparison.

4. Numerical Validation of the Reduced Model

4.1. Model, implementation, and reference calculations

The mathematical model is Equation (22); the numerical implementation is its Euler–Maruyama discretization. Each trajectory starts at $y_0 = -\sqrt{3}$ and is counted as a committed transition only on first reaching $+\sqrt{3}$. This definition matches the basin-to-basin observable in Equation (19), rather than first arrival at the saddle. Section 4 validates the canonical one-dimensional normal-form dynamics and its first-passage implementation; it does not validate the finite-μ Hopfield landscape, the stochastic dimensional reduction, or the cognitive interpretation.

The production sweep used 16 log-spaced values $r \in [0.15, 7.0]$, canonical noise $\theta = gT/\mu^2 = 3/(4r)$, $\Delta s = 0.01$, and a maximum window $s_{\max} = 150$. Exactly 6,000 trajectories were run at each point (96,000 total). The seed bases were 77,000+10,000i,

$i = 0, \ldots, 15$, and are archived with the data. The code contained a mechanism to add batches until 100 events were observed, but the initial 6,000 trajectories already exceeded this target at all points; consequently, no adaptive increase occurred in the production sweep.

Trajectories without a transition by $s_{\max}$ were right-censored. For d observed events, the constant-hazard estimate was

$$\widehat{\kappa} = \frac{d}{\sum_i t_i^{\mathrm{obs}} + n_{\mathrm{cens}} s_{\max}}. \tag{27}$$

The initial relaxation transient was handled by conditioning on survival to a candidate t^*, discarding earlier events, re-baselining the remaining times to $t - t^*$, and refitting Equation (27). Candidates were $\{0, 0.5, 1, 2, 4, 8, 16, 32\}T_{\mathrm{relax}}$, with $T_{\mathrm{relax}} = [U''(y_-)]^{-1} = 0.5$ in canonical units. The internal reference was the estimate at the largest candidate retaining at least 20 events. The selected t^* was the smallest candidate within 5% of that reference. This rule used only the empirical escape-time data, not the exact first-passage or Kramers values.

After selecting t^*, estimation was conditioned on survival to t^*. For this post-truncation set, $\widehat{\kappa} = d_{\mathrm{post}}/\mathcal{T}_{\mathrm{post}}$, where d_{post} is the number of committed transitions observed after left truncation and $\mathcal{T}_{\mathrm{post}}$ is the corresponding total time at risk, including right-censored survivors. The reported approximate 95% interval is $\widehat{\kappa}\exp(\pm 1.96/\sqrt{d_{\mathrm{post}}})$. This fit estimates a constant escape hazard over the post-t^* window conditional on survival to t^*.

For an independent one-dimensional reference, the exact first-passage expression from the backward equation was evaluated numerically. The nested quadrature follows from the standard one-dimensional backward first-passage formulation, specialized here to the present drift, diffusion, and reflecting/absorbing boundary conventions (Szabo et al., 1980):

$$\tau(y_0 \to b) = \frac{1}{\theta}\int_{y_0}^{b} e^{\mathcal{U}(v)/\theta}\left[\int_a^v e^{-\mathcal{U}(z)/\theta}\,dz\right]dv, \tag{28}$$

where $\mathcal{U}(y) = y^4/12 - y^2/2$, with $y_0 = -\sqrt{3}$, absorbing $b = +\sqrt{3}$, and $a = -20$ as a numerical representation of the confining boundary at $-\infty$. We denote the inverse point-start MFPT by $\kappa_{\mathrm{exact}} = 1/\tau$. The mean includes the initial transient of the complete first-passage distribution. Close agreement with the stabilized hazard is expected when local relaxation is rapid relative to escape and first-passage times are approximately exponential. The MFPT expression is exact within the reduced one-dimensional boundary-value problem, apart from the separately checked finite grid and cutoff; it is not an exact solution of the full Hopfield network. The Kramers curve, Equation (23), is a distinct high-barrier asymptotic approximation.

4.2. Convergence and statistical safeguards

At $r = 3.5$, varying the quadrature grid from 5,000 to 80,000 points and the lower cutoff from -15 to -30 left $\kappa_{\mathrm{exact}} = 0.005829$ unchanged to six significant figures (variation below 0.01%). The archived Euler–Maruyama step-size check used $\Delta s = 0.02, 0.01, 0.005$ at $r \approx 1, 3.5, 7$. Only a summary was retained for the two lower-barrier points. At $r = 7$, a re-test with approximately 150 events per step size gave $\widehat{\kappa} = 0.000205$, 0.000198, and 0.000183, respectively; the 3–8% changes were comparable to the approximately 8% relative Monte Carlo standard errors. These checks were limited to the three representative points.

As an additional censoring-window sensitivity check, doubling the maximum canonical observation window from $s_{\max} = 150$ to 300 changed the stabilized rate estimates by approximately 1.5%, 1.8%, 4.7%, and 5.5% at $r \approx 3.25, 4.19, 5.42$, and 7.0, respectively. Sensitivity was therefore modest but nonzero and increased toward the most strongly censored high-barrier conditions.

As a diagnostic of the observable definition, the empirical $\kappa_{\mathrm{hit}}/\kappa_{\mathrm{commit}}$ ratios at $r = 1, 2, 3.5, 5, 7$ were $2.41, 2.12, 2.00, 2.03, 2.09$, respectively, approaching the asymptotic factor of two in Equation (26). This isolates the former saddle-hit discrepancy as an observable mismatch rather than evidence for an additional dynamical correction.

4.3. Results

Figure 1 shows the three-way comparison. For $r \geq 2.5$, the stabilized simulation estimate and inverse MFPT differ by approximately 0.5–3%, and the inverse MFPT lies within the reported simulation interval at all five tested points in this regime. At lower barriers, the stabilized estimate is systematically higher than the point-start inverse MFPT. With appreciably non-exponential first-passage times, these are different estimands, so their discrepancy does not measure integration error. Kramers' expression is poor at low barriers, captures the scale at intermediate r, and approaches the exact rate progressively at high r. At $r = 7$, the stabilized simulation estimate is 0.000193 (95% CI $[0.000166, 0.000225]$), the exact first-passage rate is 0.000192, and the Kramers rate is 0.000205. Thus Kramers' rate formula emerges progressively as the high-barrier asymptotic limit of the exact first-passage dynamics, with a residual, shrinking discrepancy at the r values tested here.

The log-ratio $\log(\kappa_{\mathrm{exact}}/\kappa_K)$ decreases in magnitude to approximately -0.066 at $r = 7$. A post hoc regression over $r \geq 1.5$ was consistent with an

$O(1/r)$ correction ($R^2 = 0.86$) but had a nonzero intercept of approximately -0.046; this pattern is descriptive and does not establish the correction order. The hazard procedure selected no truncation at several moderate-to-high barrier points, where the stabilized and untruncated estimates coincide by construction. Their relative difference was 0.34% at $r = 7$.

Complete numerical values for all 16 sweep points are reported in Supplementary Table S1.

5. Discussion

5.1. Interpretation of the critical regime

Within the truncated normal form, increasing ε toward $\varepsilon_{\text{crit}}$ decreases the separation between two energetically equivalent effective states along the contrastive coordinate. At $\varepsilon_{\text{crit}}$ they coalesce; above it, the potential is single-welled at the quiescent state on that coordinate. This does not predict that insight events become impossible or that their rate collapses. Labelling the initially occupied well "impasse-like" and the opposite well "solution-like" is a modelling interpretation tied to the initial condition, not an energetic preference derived from the model. Once the two-basin normal form disappears, Kramers' basin-to-basin observable no longer applies.

The earlier numerical estimate of an "optimal" value of ε was obtained from a superseded simulation and has not been re-estimated under the corrected barrier law and committed-transition definition. The present analysis therefore supports no numerical optimum. More broadly, neither ε nor $\varepsilon_{\text{crit}}$ has been calibrated against human behavior or neurobiology.

5.2. Clinical implications as hypotheses

Possible relations between representational rigidity, associative looseness, atypical representational dynamics, and clinical symptoms remain hypotheses for future study. The reduced model supplies no diagnostic mapping, neurobiological measurement, or treatment prediction, and it does not show that any psychiatric condition has a particular W_{AB} structure or that an intervention changes ΔE in a patient. The merger of wells in the toy potential is a loss of basin distinguishability, not an established clinical mechanism. Testing any diagnosis-specific proposal would require human data, independently operationalized estimates of connectivity and TFI, preregistered behavioral outcomes, and clinical safeguards.

5.3. Reconciling RCT and the Free Energy Principle

RCT (Ohlsson, 1992, 2011) describes representational operations associated with insight, whereas RFTIC proposes a stochastic landscape in which such a transition might be represented. This is a conceptual correspondence, not a confirmation or formal derivation of RCT. Likewise, the model parameter T is a noise amplitude. Relating it to precision weighting or dopaminergic modulation in the Free Energy Principle (Friston, 2010; Friston et al., 2017) is a possible interpretation, not a result derived or tested here. Empirical work must determine whether ε and T can be estimated separately and whether they predict human performance as the reduced model suggests.

5.4. RFTIC for Artificial General Intelligence

Changes in learned representations in artificial systems are only analogous to modification of W, and inference-time sampling temperature is only loosely analogous to the noise parameter T; no mathematical identity has been established. Cross-domain fine-tuning, regularization, and self-consistency procedures are possible experimental manipulations, but ARC or ARC-like evaluation is not presently predicted by a measured TFI, and no artificial-system validation is reported here (Chollet, 2019). Such comparisons belong to future work requiring explicit operational definitions, controlled baselines, and measurements that separate learned representation change from inference-time sampling.

5.5. Limitations and Future Directions

The one-dimensional reduction of the simplified continuous Hopfield energy rests on orthonormal ψ_A, ψ_B, the baseline eigendirection hypothesis (A1), and the transverse-gap hypothesis (A2). A2 is an additional sufficient condition for transverse convexity; it is not empirically established for real semantic networks and excludes some natural symmetric two-pattern storage matrices. The exact eigenvalue shift applies to the specified Hebbian perturbation, but does not by itself justify the one-dimensional reduction.

The reduced energy is an asymptotic normal form near $\mu \to 0^+$. Its Level-B model truncates the saturation expansion at quartic order and neglects $O(x^6)$ corrections; the leading relative barrier correction is $-36c_6\mu/g^2 + O(\mu^2)$. Its validity also depends on $g = \sum_i (v_i^*)^4$: the barrier scales as $1/g$, while the effective canonical noise contains gT. The wells are effective contrastive states, not the original patterns ψ_A and ψ_B, and the impasse/solution labels remain interpretive.

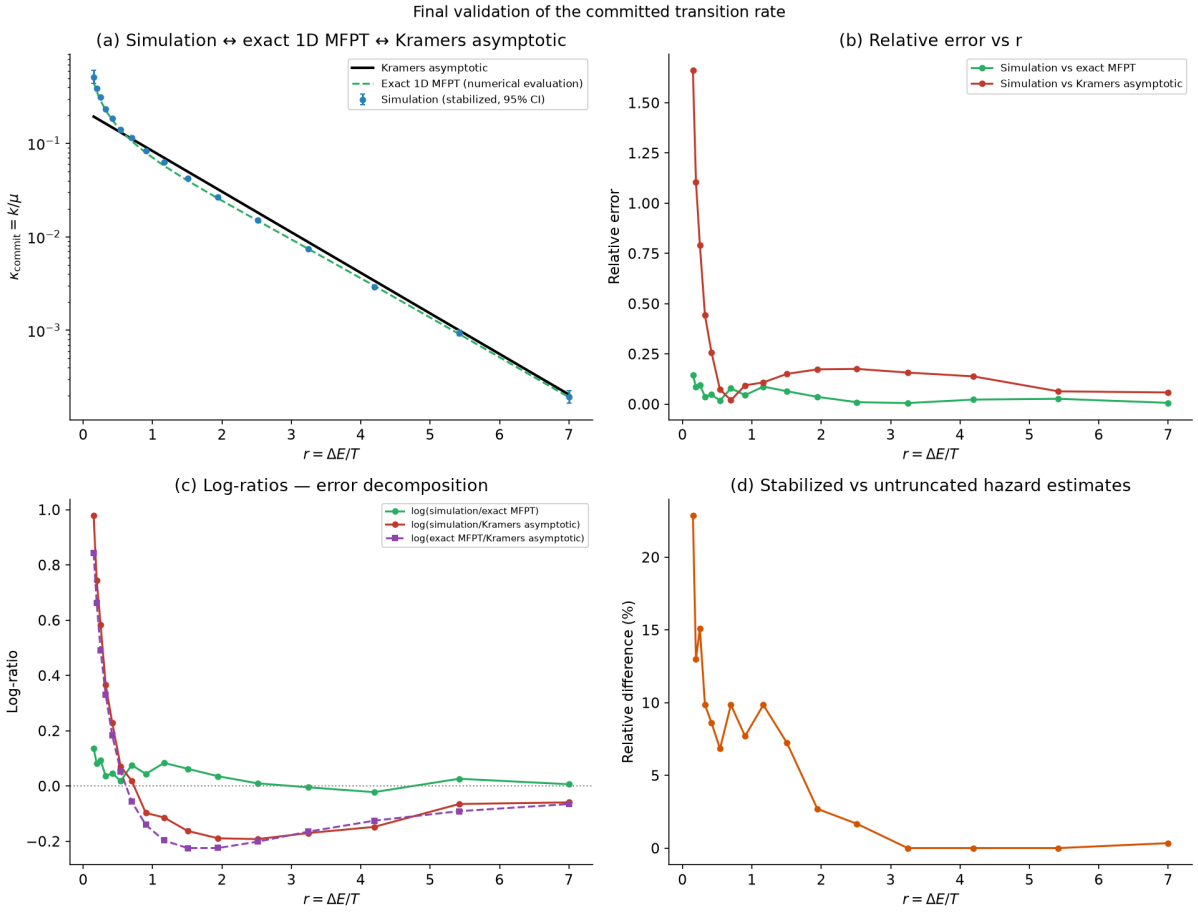


Figure 1: Validation of the committed basin-to-basin rate across $r = \Delta E/T$. (a) Hazard-stabilized simulation estimates with right-censoring and approximate 95% confidence intervals, the numerically evaluated inverse point-start MFPT, and the Kramers asymptotic formula. (b) Absolute relative differences between the stabilized estimate and each reference. At low barriers the stabilized hazard and inverse MFPT target distinct quantities. (c) Log-ratios separating the stabilized–MFPT and MFPT–Kramers differences. (d) Absolute difference between the stabilized and untruncated hazard estimates, expressed as a percentage of the untruncated estimate.

Reducing the energy by transverse partial minimization is distinct from reducing the stochastic dynamics. The adopted one-dimensional Langevin equation assumes isotropic additive noise and unit mobility, and adiabatic slaving of transverse variables (requiring $\mu \ll \gamma$). The value $\varepsilon_{\text{crit}} = \mu_0$ is the deterministic critical point of the adopted zero-noise energy reduction. In a full-dimensional Langevin system, transverse stochastic fluctuations may renormalize the effective reduced coefficients or change the apparent critical behavior. These corrections were not quantified, so the deterministic critical point does not establish a noise-independent cognitive threshold. No full-dimensional Hopfield–Langevin validation was performed, and a two-dimensional extension is outside the present analysis.

The numerical results show close agreement between the stabilized hazard and inverse MFPT at moderate-to-high barriers. Kramers' expression remains asymptotic; its leading correction order was not derived, and the tested range retains a residual discrepancy. Euler–Maruyama convergence was checked at only three r values. The production confidence intervals use an asymptotic log-normal approximation with the post-truncation event count. They condition on the empirically selected t^* and do not propagate uncertainty in the choice of truncation point. Finite-window sensitivity was nonzero and reached approximately 5.5% at $r = 7$ when $s_{\max}$ was doubled, so the present results do not establish complete independence from $s_{\max}$. No human behavior, neural, clinical, or artificial-intelligence data were analyzed. Future work should test the reduction in full-dimensional networks, quantify departures from A1–A2 and orthonormality, and evaluate the proposed cognitive variables with preregistered human studies.

5.6. Conclusion

RFTIC offers a testable distinction between a proposed structural contribution to representational change and transient state-dependent noise. Under its explicit reduced-model assumptions, the cross-domain perturbation shifts the raw transition-mode eigenvalue by $-\varepsilon$. Under A1–A2, the Hopfield energy then has a near-bifurcation asymptotic normal form with $\mu = \lambda_W - 1$, quartic coefficient g, Level-B barrier $\Delta E = 3(\mu_0 - \varepsilon)^2/(4g)$, and critical point $\varepsilon_{\text{crit}} = \mu_0 = \lambda_{W0} - 1$. At moderate-to-high barriers the stabilized simulation hazard agrees closely with the inverse point-start MFPT. The Kramers expression approaches the inverse MFPT progressively at high barriers.

These conclusions concern mathematical and computational consistency within the reduced model. Whether representational flexibility is a measurable, trainable human capacity, whether its proposed components alter semantic connectivity, and whether the framework has clinical or engineering value remain open empirical questions.

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Conflict of Interest

The author declares no conflict of interest.

Author Contributions

M. Marmore conceived the RFTIC framework and research questions, initiated the project independently, defined the modeling objectives and theoretical assumptions, directed the mathematical and computational development, designed the validation strategy, evaluated the derivations and simulations, interpreted the results, and determined the final scientific claims and conclusions of the manuscript.

M. Marmore is responsible for the final scientific content of the work.

Declaration of AI Tool Use

Generative artificial-intelligence systems, including ChatGPT (OpenAI) and Claude (Anthropic), were used as research-support tools during the development of this work.

Their use included assistance in understanding relevant literature and mathematical, physical, statistical, and computational concepts; assistance in developing, reconstructing, and checking selected mathematical derivations; critical review aimed at identifying possible inconsistencies; assistance in developing, reviewing, and debugging simulation and analysis code; support in evaluating numerical validation procedures and intermediate computational results; and language, grammar, and structural revision of the manuscript.

AI-generated suggestions were treated as provisional research inputs rather than as authoritative results. Mathematical and computational claims

considered for inclusion were independently examined through analytical re-derivation, symbolic or numerical computation, internal consistency checks, and comparison with the governing equations.

The research questions, theoretical framework, modeling assumptions, decisions regarding the acceptance or rejection of proposed derivations, interpretation of results, and final scientific claims were determined by the author.

Ethics Statement

This study did not involve human participants, animals, identifiable personal data, or clinical intervention. The present work is theoretical and computational.

Data and Code Availability

The simulation and analysis code, production sweep data, fixed seed registry, per-trajectory escape-time data, validation figure, and reproducibility materials are publicly available at `https://github.com/Khaos41x/rftic-insight-model`.

The production validation sweep used 16 values of the dimensionless barrier ratio $r = \Delta E/T$, with 6,000 simulated trajectories per condition.

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